

(ECS708) Report lab4: Classification

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Q1 - We again notice that the attributes are on different scales. Use the normalisation method from last lab, to standardize the scales of each attribute on both sets. Plot the normalized and raw training sets; what do you observe?

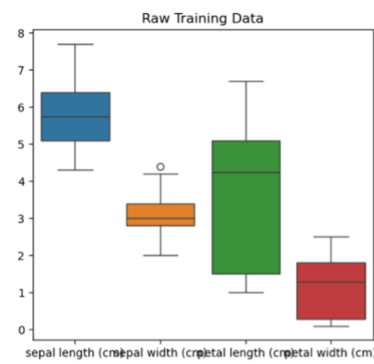


Figure 1: Boxplot raw training data

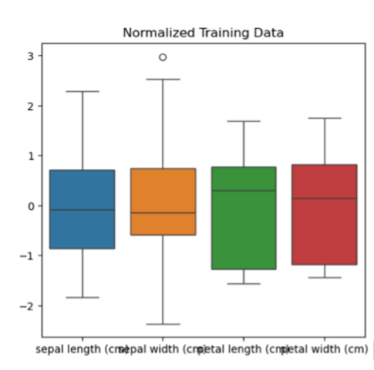


Figure 2: Boxplot normalized training data

Before normalization, each feature was on a different scale, as shown in Figure 1:

- Sepal length ranged between 4-8
- Sepal width ranged between 2-4
- Petal length ranged between 1-7
- Petal width ranged between 0-3

Using the same normalization method as in the previous lab (z-score normalization), all features align more closely around the same mean (0) and have a similar scale, allowing the classifier to treat each feature with the same importance (Figure 2).

Q5 - Draw the decision boundary on the test set using the learned parameters. Is this decision boundary separating the classes? Does this match our expectations?

The decision boundary learned by the logistic regression model forms a straight line that separates Setosa samples from the other two classes in the test set, as shown in Figure 3. Points on one side of the line are classified as Setosa, while points on the other side are classified as not Setosa (classes 1 and 2).

In this case, the decision boundary successfully separates the classes, which means the model correctly learned a linear relationship that distinguishes Setosa from the others based on sepal length and width.

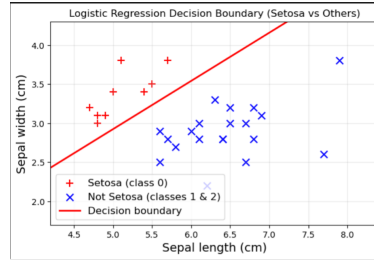


Figure 3: Logistic regression decision boundary (setosa vs others)

This was expected and matches our expectations, as in the first initial distribution visualization (Figure 4), it was already apparent that Setosa was linearly separable from the other two classes. Whereas classes 1 and 2 feature overlap, making them harder to separate linearly.

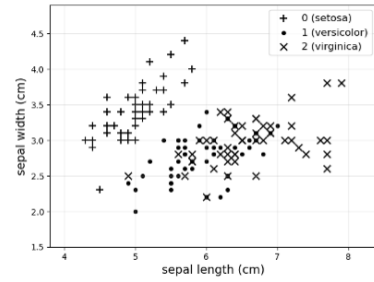


Figure 4: Distribution of Iris classes based on sepal width and length

Q7 - Using the 3 classifiers, predict the classes of the samples in the test set and show the predictions in a table. Do you observe anything interesting?

Using the three trained one-vs-all classifiers and combining their outputs, I predicted the classes of the test set samples, where the predicted class corresponds to the index with the maximum probability.

By observing the first 20 samples (Table 1), all are correctly classified. This indicates that the model generalizes extremely well to unseen data. As noted earlier, Setosa is easily linearly separable, resulting in perfect classification performance for this class. Minor misclassifications would most likely occur between Versicolor and Virginica due to their overlapping features, but it hasn't been the case.

True	Predicted
1	1
0	0
2	2
1	1
1	1
0	0
1	1
2	2
1	1
1	1

Table 1: Ground truth vs Prediction (10 examples)

Q8 - Calculate the accuracy of the classifier on the test set, by comparing the predicted values against the ground truth. Use a softmax for the classifier outputs.

Applying the softmax to the classifier outputs ensures that the class probabilities are normalized

between $[0,1]$ and sum 1. The resulting test accuracy is 1.0 (100%), which means the model correctly classified all test samples.

Q9 - Looking at the datapoints below, can we draw a decision boundary using Logistic Regression

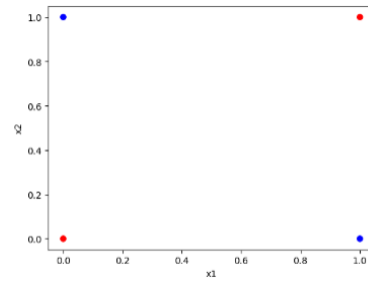


Figure 5: XOR pattern

In this pattern, the red points are at (0,0) and (1,1), while the blue points are at (0,1) and (1,0), as shown in Figure 5. In this configuration, no single straight line can separate the red from the blue, as the classes are arranged diagonally. Therefore, we can't use logistic regression, as it is a linear classifier that can only learn linear decision boundaries.

So, this XOR pattern shows one of the limitations of logistic regression, that it cannot solve problems that require nonlinear separation. No combination of weights and bias can make a single sigmoid boundary that correctly classifies all points.